Beehive AI monitoring and robotic automation



Artjom Kurapov
Founding engineer

GRAHEON

Precise pollination

- Pollination boosts crop yields (+10-30%)
- Farmers lack sufficient amount of pollinators
- Beekeepers providing services to farmers earn 9x more money compared to their honey income
- Demand of pollination grows 2x faster than growth of honeybee colonies



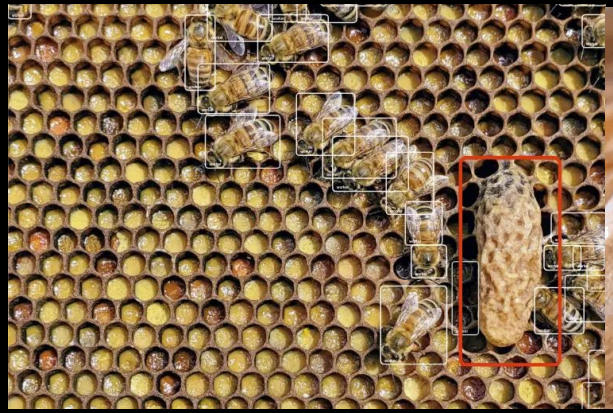
Human worker efficiency

- Hard physical labour
- Not scalable with amount of beehives
- Driving to remote locations takes time



Bee colony health observability

- Need regular (weekly) inspections
- Treat against varroa mite infestations
- Predict swarming, track queen presence
- Track colony growth/efficiency, prevent starvation





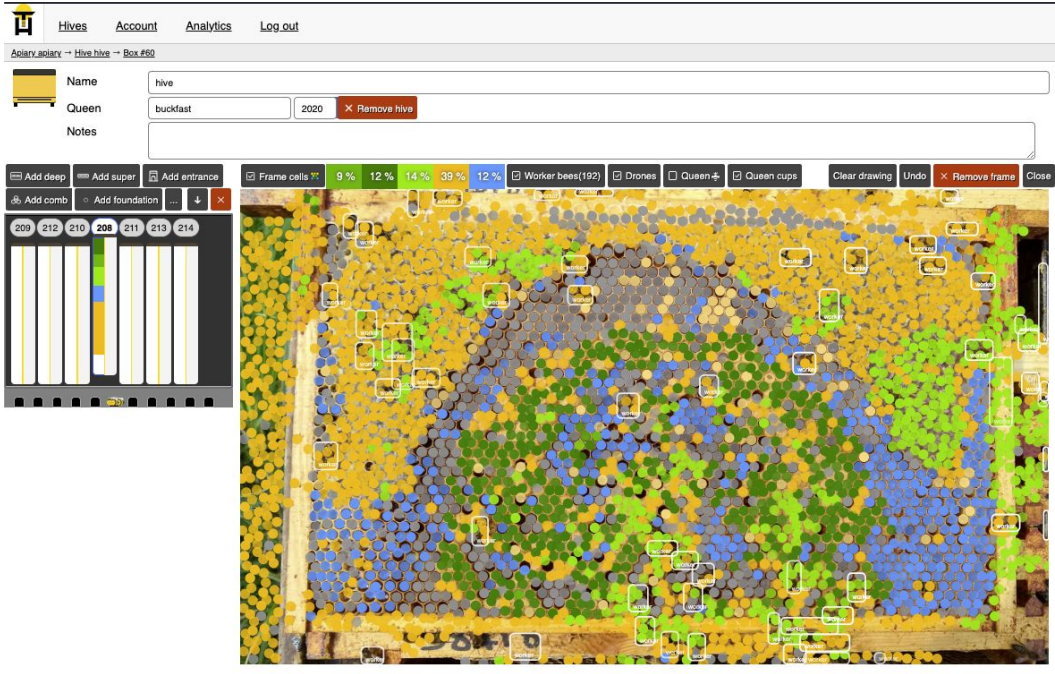
Vision

app

Software as a service for beekeepers to organize and monitor their apiary

- ☑ AI detections of a frame photo
- ☑ manage apiaries, hives, frames, photos
- ☑ Queen and queen cups
- ☑ Varroa mites
- ☑ Frame cells
- ☑ AI advisor
- ☑ Mobile app
- ☑ Manual inspections

- ▢ Re-train on user data
- ▢ Analytics (graphs) to find correlations/anomalies
- ▢ Alerts



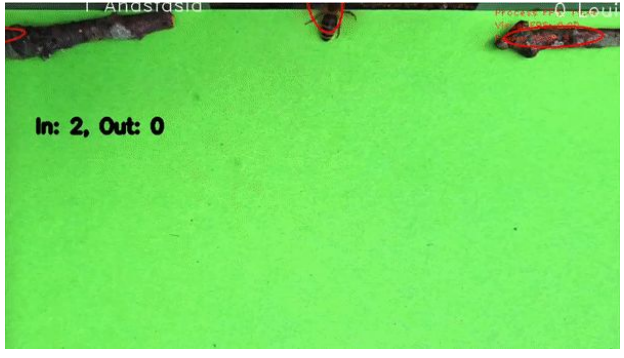
Vision

app → Entrance Observer

Hive entrance video monitoring / IoT device

- ☑ Video streaming & playback
- ☑ Incoming/Outgoing bee counter

- ▢ Varroa mite detection
- ▢ Foraging analysis
- ▢ Stealing state
- ▢ Hornet detection



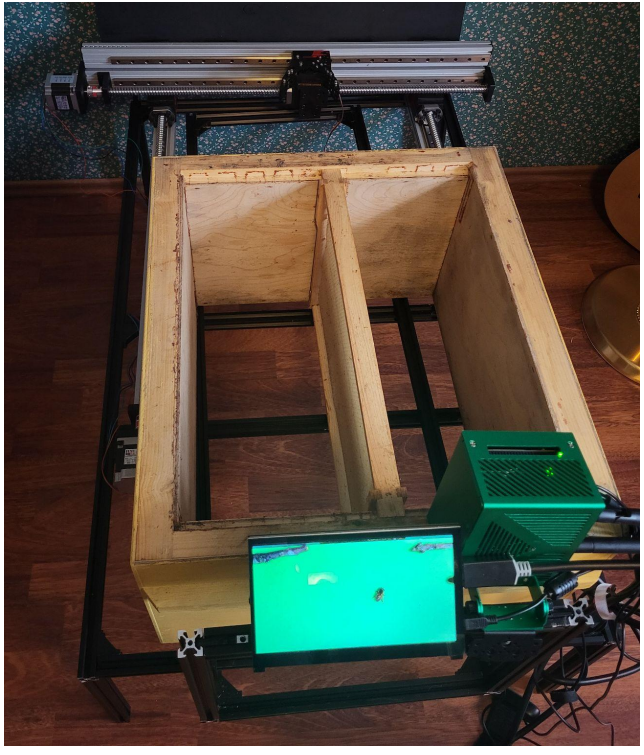


Vision

app → Entrance Observer → Robot

- Frame extraction mechanism
- 2 cameras
- Robotic beehive. Initially single colony
- Later, moving on rails for inspect multiple hives
- Permanent frame extraction
- Frame movement across hives
- Ventilation control

- Move robot on wheels to serve multiple beehives



early prototype

Team



Artjom Kurapov
Founding engineer / beekeeper
(ex-Pipedrive, Clarifai)



Muhammad Zain Shakeel
Mechanical engineer volunteer



Aleksei Zaitsev
Fullstack engineer volunteer



Ahmed Daoudi
Fullstack engineer volunteer

Research advisors, Czech Republic



Šimon Bilík
Researcher, PHD
System engineer / Beekeeper



Adam Ligocki
ML engineer, PHD

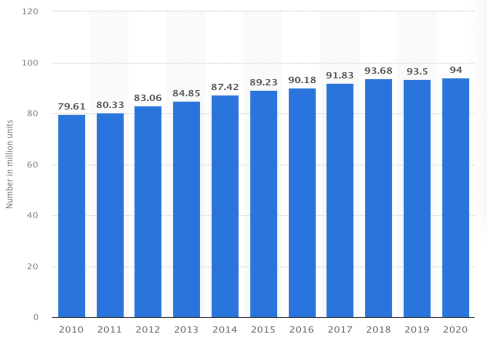
Pricing model

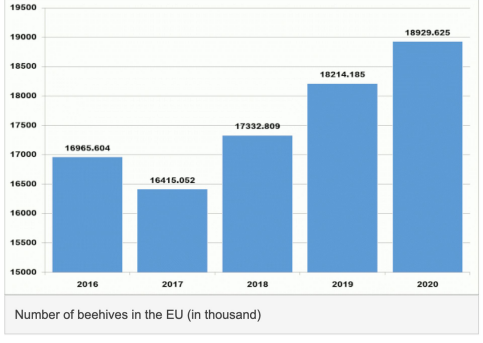
Community	Essential	Professional
free	**15** EUR / month 2 weeks trial, annual billing	**5** EUR per beehive per month + **10** EUR per user per month
5 hives max	✅ More AI detection features ✅ Timeline ✅ Sharing ✅ Alerting ✅ Telemetry API	(All of Essential plan, plus) ✅ Analytics + Demographic breakdown ✅ Multiple users ✅ Inventory management ✅ Video streaming playback

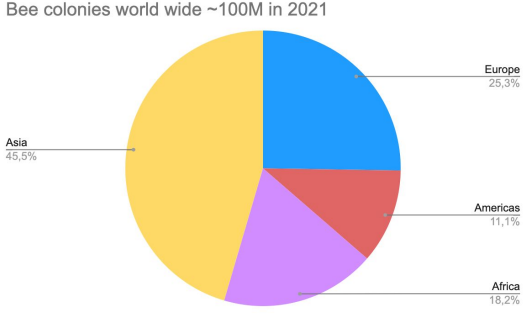


Market

- Target customers - **industrial beekeepers** (B2B, robot)
- Secondary - backyard beekeepers (B2C, web-app, manual inspections)
- Addressable market
  - Estonia - 6425 **apiaries** (PRIA sources). ~ 60% beekeepers have > 25 bee colonies
  - Europe - 620k **beekeepers, 19-25M** colonies
  - World wide - 94M **colonies** in 2020 → 101.6 M **colonies** in 2021







Market

- 10% market share * 1M industrial beekeepers * 2 robots * 150 EUR/mo = 360M EUR YoY
- 5% market share * 2M beekeepers * 15 EUR/mo for web-app = 18M EUR YoY

Go to market strategy by Product development

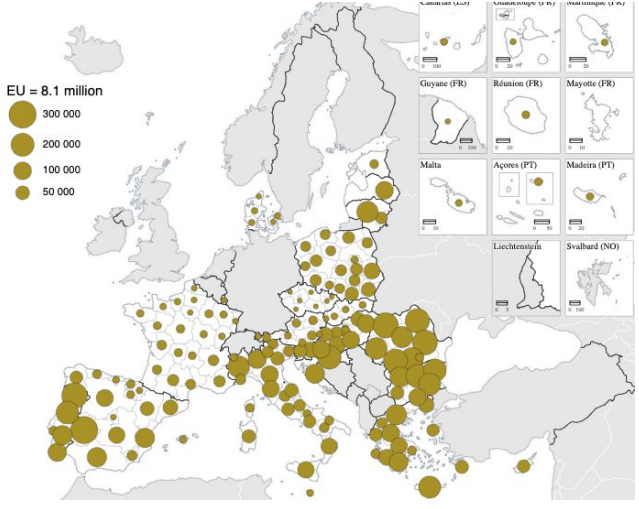
- Web-app
- Entrance observer
- Robotic beehive + apiary

Go to market strategy by region

- Estonia, Baltics (Seed / R&D phase)

- Europe - Poland, Hungary, Croatia, Italy, Bulgaria (Series A)

- US, Middle East, Portugal (Series B)



Number of beehives on farms, 2020

Invest

- Raising 35k angel investments
  - Hardware development (Entrance observer)
  - Field testing with local beekeepers
  - GPU hosting cluster + cloud video storage

- Raising 1M pre-seed round for runway: 24 months
  - **Team of 4** + external contractors & beekeepers
  - AI models improvements
  - Robot R&D

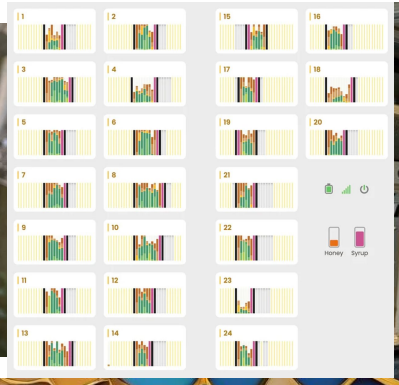
pilot@gratheon.com

Competition – Vision AI and hardware

- beewise.ag - robotic multi-colony container hive
- beehero.io - IoT
- beemate.buzz - counts bees
- apic.ai
- bestbees.com





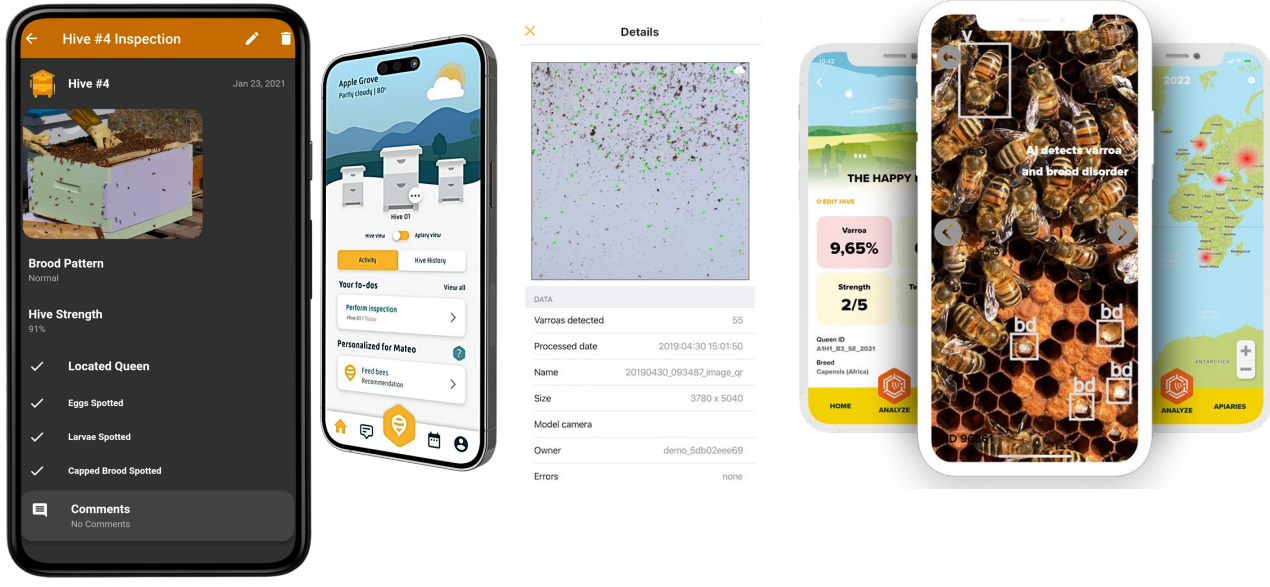






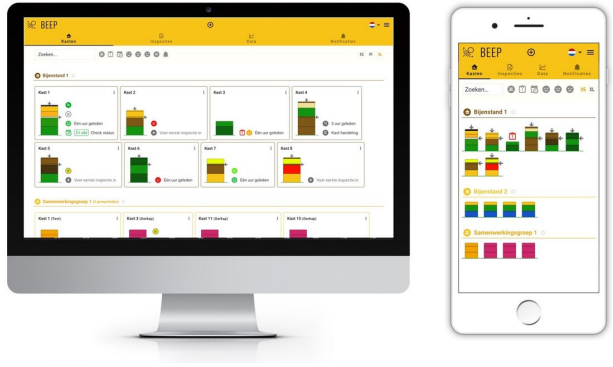
Competition - Data organizer apps

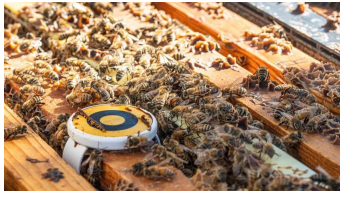
- nectar.buzz
- BeeScanning
- ApiZoom
- HiveTracks
- HiveBloom
- BeeQueenDetector
- apimanager
- apiary book

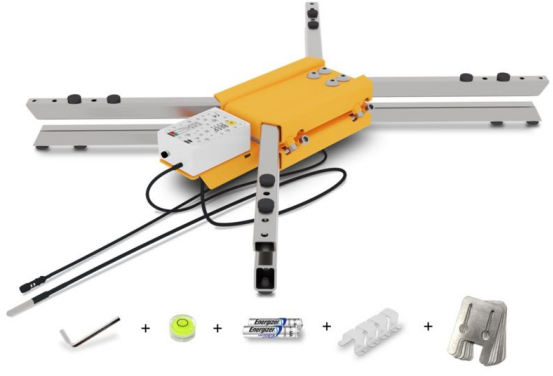


IoT – analog data (audio, humidity, temperature)

- beehero.io
- beep.nl - opensource
- broodminder.com
- beelab.se
- intelligenthives.eu
- beehivemonitoring.com
- solutionbee.com
- beehivemonitoringusa.com
- osbeehives.com
- beesage.co







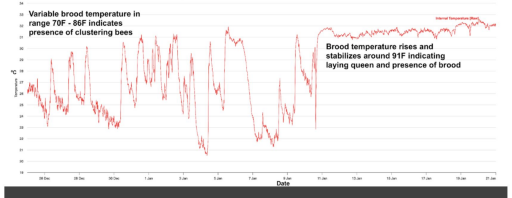


Fig. 2: Using Brood temperature to detect onset of laying queen in late winter/early spring